Hardware for a Construction-Site Digital Twin

*Research date: 2026-10-18. Price legend: **(~)** = reseller, third-party or estimate; **(?)** = sources conflict or unverified. Figures without a URL are background knowledge, not verified in this session.*

1. Capture sensors

1a. LiDAR sensors you can mount on a robot or vehicle

Model	Key specs	Price (2025-26)	Notes
Unitree L2	360°×96° FoV, 64k pts/s effective, 30 m range, ≤2 cm accuracy, built-in IMU, 230 g	**$419** (Unitree)	Cheapest usable 3D LiDAR. Good for SLAM; too noisy for QA.
Livox Mid-360	360°×59°, ~200k pts/s, 40 m at 10% reflectivity, ≤2 cm at 10 m, built-in IMU	$770-1,250 (China sellers); $3,800 at US resellers (RoboStore)	Standard sensor for FAST-LIO. An ISPRS 2025 test of 35 units found all within spec (paper).
Livox Avia	70°×77° FoV, 450 m range	$2,079 (DJI)	Suits drones and long range.
Hesai JT128	128 channels, 360°×189° hemispherical	$1,129-1,500 (SensorPartners)	Excellent value. Sees floor and ceiling in one pass.
Hesai XT32 / XT32M	32 channels, 120 m range, ±1 cm	$3.7-4.2k (XT32M ~$5.4-6.2k) (LidarSeek)	Survey-grade mobile LiDAR core.
Ouster OS0 / OS1	OS0: 90° vertical FoV, ~50 m. OS1: 45° vertical FoV, ~120 m. 32-128 channels	OS1-32 ~$6.4k; OS1-128 ~$12k **(~)**, sold by quote	Best ROS drivers. Also outputs near-IR and ambient images.
Velodyne VLP-16	Legacy	used $1-3k **(~)**	End-of-life.

1b. Terrestrial laser scanners (tripod, survey-grade)

Model	Accuracy	Price
Leica BLK360 G2	~4 mm at 10 m	~$28.7k **(~)**
Leica RTC360	1.9 mm at 10 m	$65-100k new **(~)**
FARO Focus Premium	±1 mm ranging	~€38k EU; $66-84k US **(?)**
Trimble X7	2.4 mm at 10 m	~$35-50k **(~)**

1c. Mobile and handheld SLAM scanners

Model	Accuracy	Price
Leica BLK2GO	6-10 mm relative	$37.5-61k (Leica)
FARO Orbis	5 mm SLAM, 2 mm flash mode	~$20-35k **(~)**
GeoSLAM/FARO ZEB Horizon RT	1-3 cm	~$17-26k **(~)**
NavVis VLX 2/3 (wearable)	~5 mm (claimed)	€60k to $130-170k **(?)**
Emesent Hovermap ST-X	1-2 cm; works on drone, Spot, vehicle or backpack	$80-120k **(?)**

Software adds $5-15k per year (Cyclone, IVION and similar).

1d. Cameras

Type	Model	Price
360°	Insta360 X5 (72 MP stills, 8K30 video)	~$465
360°	Ricoh Theta Z1 (standard camera for OpenSpace-style walkthroughs)	~$800-1,100 **(~)**
Stereo	ZED X (GMSL2, IP67, global shutter, built for Jetson)	$599
Stereo	ZED 2i (USB)	$499
RGB-D	RealSense D455 (noisy outdoors)	~$300-420
Stereo + AI	OAK-D Pro W (runs inference on-device)	~$350-400 **(~)**
Thermal	FLIR Lepton/Boson	$250 to $3k+ **(~)**

1e. Drones

Model	Use	Price
DJI Matrice 4E (replaces the Mavic 3E)	RTK photogrammetry, 1-3 cm with ground control	~$5.1k
DJI M400 + Zenmuse L3	LiDAR, 3-4 cm RMSE	~$28k
Skydio X10	Autonomous, NDAA-compliant	$15-25k by quote
DJI Dock 3 + M4D	Drone-in-a-box for daily automated flights	from $15.9k

US restrictions on DJI (NDAA and FCC rules) are a procurement risk. Skydio is the compliant alternative.

1f. GNSS RTK and IMUs

- **Emlid Reach RS3/RS4:** $3-4.6k each; base-plus-rover kit $6-8.9k. Accuracy ~7 mm + 1 ppm.
- **Robot-mounted RTK:** u-blox ZED-F9P (~$200-300) or Unicore UM980 (~$150-250).

- **IMUs:** the built-in Livox/Ouster IMUs are enough for LIO. Xsens MTi-680G costs ~$2-5k.

2. Persistent site sensors for a "live" twin

Category	Examples	Price	Notes
Fixed / PTZ cameras (managed)	OxBlue, EarthCam, TrueLook	$2-20k hardware + $200-900/mo	Time-lapse and progress.
DIY IP cameras	Axis, Reolink with 4G and solar	$150-2,000 per camera; solar trailer $5-15k	Hikvision and Dahua are NDAA-restricted.
Crane cameras	Versatile, EarthCam	Quote only	Near-daily top-down views without a drone.
UWB / BLE tags	UbiTrack, Pozyx, Sewio, Redpoint	Tags $25-99; anchors $499-799	UWB 10-30 cm; BLE 1-5 m.
Equipment telematics	ISO 15143-3 (AEMP 2.0): Cat, Komatsu, Volvo, Deere	Included with OEM subscription; retrofit $100-300 + $15-30/mo	Location, hours, fuel, idle time, fault codes. Updates every 10 minutes to 24 hours.
Concrete maturity	Giatec SmartRock	$135-250 per sensor	
Dust, noise, vibration	Aeroqual, Kunak; Svantek, Sigicom	$250 to $10k	Permit compliance.
Structural	Tiltmeters, crack meters, automated total stations	$300-1,500 per node; total station $60k+	Excavations and neighbouring buildings.
Connectivity	LoRaWAN gateway $300-1,500; private 5G $20-100k+ **(?)**; Starlink $1,999 + ~$250-290/mo		Scan uploads need Starlink, 5G or carrying an SSD off site.

3. Capture platforms

Platform	Cost	Strengths and weaknesses
Handheld / pole	$1-60k	Cheap and flexible, but labor-heavy.
Backpack / helmet	$5-170k	Fast coverage of floors.
Wheeled AMR (the opening for Flo Mobility)	$10-40k custom	Repeatable routes, cheap per capture. Struggles with stairs, rubble and mud.
Legged (Spot $150-375k; Unitree B2 ~$77-106k; Go2 EDU ~$10-15k)		Stairs and rough ground. Spot + BLK ARC is the reference integration.

Platform	Cost	Strengths and weaknesses
Drone	$5-30k	Exteriors and earthworks. No interiors once the building is enclosed.
Vehicle mobile mapping	$30k-500k	Roads and large sites.

4. Compute and storage

Edge

Module	Price	What it can run
Jetson Orin Nano Super	$249	FAST-LIO2 + light YOLO + logging
Jetson Orin NX 16 GB	~$999 module	LIO + stereo + 2-4 camera detection + nvblox
Jetson AGX Orin 64 GB	$3,499	Several LiDARs + 4-6 cameras + on-board change detection
Jetson AGX Thor	$5,499	VLMs on-device **(?)**

NVIDIA raised Jetson prices in 2026.

Data rates: a rosbag with LiDAR + IMU + 2 compressed cameras is ~0.5-2 GB per minute, so plan 2-4 TB of NVMe on the robot.

Reconstruction

- **Photogrammetry:** 1-3k images need 64-128 GB RAM and a 24 GB GPU, taking 2-8 hours.
- **Gaussian splatting:** a medium scene needs ~24 GB VRAM; a large scene 40-80 GB. An RTX 4090 trains 30k iterations in ~30-40 minutes. CLM-GS fits 100M+ Gaussians on a 24 GB GPU.
- **Recommended:** a workstation with an RTX 5090 or RTX 6000-class GPU and 128-256 GB RAM ($6-15k), or cloud A100/H100 at ~$1.5-4 per GPU-hour.

Storage

A weekly capture of a medium site is ~50-200 GB raw, or about **3-10 TB per project-year**. Serve it through Potree or 3D Tiles.

5. Recommended bundles by tier

USIBD accuracy bands: **LOA10** worse than 5 cm, **LOA20** 15-50 mm, **LOA30** 5-15 mm, **LOA40** 1-5 mm. Progress tracking needs about LOA10-20 (2-5 cm). QA against BIM needs about LOA30 (1-2 cm or better).

Tier	Bundle	Cost	Accuracy	Good for
(a) MVP	Mid-360 or Unitree L2 + Insta360 X5 + F9P/UM980 RTK + Jetson Orin NX/Nano + SSD, on a pole **or a Flo robot**	$2.5-5k	1.5-3 cm relative; 2-5 cm absolute with RTK and control points	Progress tracking, visual twin, navigation maps. Not QA.

Tier	Bundle	Cost	Accuracy	Good for
(b) Mid	Hesai JT128 or Ouster OS0 + ZED X + 360° camera + Emlid RS4 + AGX Orin; plus an M4E drone; GPU workstation	$15-30k	1-2 cm relative, 2-3 cm absolute	Weekly progress, approximate deviation checks, earthwork volumes
(c) Survey	RTC360 or X7 + BLK2GO/VLX/Hovermap + M400 + L3 + Spot + BLK ARC	$80-400k+	2-6 mm	QA/QC against BIM, contractual as-builts

Suggested path for Flo Mobility

1. Build tier (b) on your own AMR, for repeatable automated captures.
2. Calibrate against a rented TLS (~$500-1,000/day) so you can report LOA bands credibly.
3. Add the live layer: cameras, UWB, telematics, maturity sensors, LoRa.
4. Reach QA through partnerships or importing data from survey scanners rather than building it.